

A self-started predictor–corrector method for calculating the Lambert W function within the scope of the photovoltaic single diode model

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ABSTRACT

The single diode model is one of the most employed to represent the behavior of photovoltaic generators. Nonetheless, the transcendental nature of its characteristic equation requires that some mathematical approach be specifically adopted to allow the determination of its solution. In recent years, the approach via the Lambert W function has gained more and more popularity; however, the calculation of this special function still faces performance issues. That said, the present paper proposes a *self-started predictor–corrector method* for calculating the Lambert W function in the interval of interest of the single diode model. This method was properly designed to return high-quality solutions at a low and fixed computational cost. In this sense, the moderate complexity of its four equations stands out, as well as its non-conditional and non-iterative nature. The results obtained for appropriate case studies ascertain its elevated level of accuracy. Also, they demonstrate that it is more effective and faster than other state-of-the-art alternatives. Therefore, the paper's major contribution is this novel method, which proves to be reliable, easy to implement, and useful for making the employment of the single diode model in studies more efficient.

1. Introduction

In recent years, *solar photovoltaic* (PV) systems have been leading the global expansion of installed capacity for electricity generation [1]. In addition to presenting an increasingly competitive cost, this kind of system brings together technical features that make it exceptionally attractive for both distributed and centralized applications [2–4]. For instance, its great modularity, emission-free operation, and reliance only on inexhaustible solar irradiation are noteworthy [5]. Nonetheless, the intermittent nature of this energy source and the strong influence of environmental factors make the efficient management of its generation a major challenge [6–8]. In view of that, accurate and fast modeling of the performance of PV generators becomes crucial for carrying out engineering activities [8–10].

Diverse performance models of PV generators are available in the literature [11–13]. Among them, the so-called *single diode model* stands out as one of the most widespread in academia and commercial software [14–16]. Classified as a physical model, it seeks to describe the main processes and attributes that define the behavior of a PV generator by establishing an equivalent electrical circuit [14]. So, through the analysis of this circuit, it is possible to obtain a parametric equation that allows the estimation of the electric current (I) produced by the PV generator under study as a function of the voltage (V) simultaneously verified at its terminals [14].

Nevertheless, the employment of the single diode model in the representation of a PV generator encounters some difficulties in the real world [14]. The primary one is that its characteristic equation has a transcendental nature [14]. In other words, the term I appears implicitly in this equation so that $I = f(V, I)$. Various mathematical approaches have already been proposed to address this issue [17]. Despite this, it is observed that the approach based on the *Lambert W function* is the one that has been gaining more popularity in the last decade. Some meaningful cases of its use can be checked in [14], [18], and [19].

Compared to other consolidated approaches, such as the one based on Newton's method, the main benefit of the Lambert W function is that it allows explicitly rewriting the characteristic equation of the single diode model [14]. Hence, it facilitates the manipulation of terms and enables the development of different analytical studies [20–22]. Notwithstanding, it is important to point out that the numerical resolution of the resulting equation of this approach depends on knowing the value of the Lambert W function itself at a point “ z ” of the non-negative real interval of its domain – that is, $I = f(V, W_0\{z\})$ [14].

Since the Lambert W function has multiple applications, several scientific programming environments offer solvers dedicated to its calculation [23–25]. The Matlab software, for example, provides a

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high-level function named “*lambertw*” [25]. Although it is widely recognized for its accuracy, some recent works have drawn attention to its relatively long simulation time [17,26,27]. So, for studies such as those related to the parameter extraction problem of the single diode model from an I - V curve, where up to thousands of evaluations of its characteristic equation may be required, using this Matlab function can be troublesome [14].

Alternatively, some self-implementing methods have been specifically developed to allow the efficient calculation of the Lambert W function within the scope of the single diode model. In this sense, the *hybrid method* introduced by [28] and the *explicit method* presented in [27] stand out as non-iterative options. Additionally, the *numeric method*, also presented in [27], emerges as a qualified iterative option. The results reported by [17], [26], and [27] demonstrate that the hybrid and explicit methods can return solutions for the Lambert W function much faster than the corresponding Matlab function; however, this comes at the expense of a loss of accuracy. In contrast, they reveal that the numeric method can return solutions identical to the corresponding Matlab function in an intermediate simulation time [27].

Given these considerations, the present paper proposes a *self-started predictor-corrector method* for calculating the Lambert W function within the scope of the single diode model. By making use of the knowledge available in specific literature, this method was fundamentally designed to return high-quality solutions in a non-iterative manner. Furthermore, its plainness of implementation compared to other related methods is remarkable. Accordingly, the most expressive contributions of this paper are three-fold:

- A self-started predictor-corrector method is innovatively proposed to allow accurate and efficient calculation of the Lambert W function within the scope of the single diode model.
- The results obtained from six appropriate case studies prove that it accomplishes its goal and corroborate the pertinence of its development.
- These results also ascertain that it is capable of returning solutions identical to those of the Matlab function “*lambertw*”, being more effective than the hybrid and explicit methods and faster than the numeric method.

In addition to this introductory section, the rest of the paper is organized as follows. Section 2 provides a brief description of the single diode model, as well as its approach via the Lambert W function. Section 3 delivers an overview of the methods already developed for calculating the Lambert W function in this scope and then details the proposed method. Section 4 presents the results obtained in the case studies used for its validation. Section 5 concludes this paper.

2. Problem statement

This section initially provides a brief description of the single diode model. In sequence, its approach via the Lambert W function, as well as the challenge behind it, are duly explained.

2.1. Single diode model

As depicted in Fig. 1, the *single diode model* is characterized by an electrical circuit composed of a current source antiparallel with a diode (D), a shunt resistor, and a series resistor. Each one of these components represents a fundamental mechanism that defines the electrical behavior of a PV cell and, thereby, its performance [5].

As the PV cell is the elementary device of a PV module and, consequently, a PV array, the single diode model can be properly employed to represent any PV generator [29]. By applying the Shockley diode equation and Kirchhoff's Laws in the analysis of the resultant circuit, it is found that the electric current “ I ” produced by a PV generator under

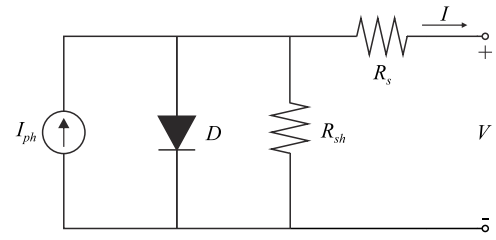


Fig. 1. Equivalent electrical circuit of the single diode model.

study can be calculated from the voltage “ V ” simultaneously verified at its terminals through (1) [14,27].

$$I = I_{ph} - I_0 \left[\exp \left(\frac{V + IR_s}{a} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \quad (1a)$$

$$a = \frac{N_s n k T_c}{q} \quad (1b)$$

In (1a), I_{ph} is the *photocurrent* of the PV generator, I_0 is the *reverse saturation current*, R_{sh} is the *shunt resistance*, and R_s is the *series resistance*. The auxiliary term a , in turn, is defined as given (1b), where n is the *ideality factor* of its cells (dimensionless), T_c is the operating temperature of them (given in kelvin), and N_s is the number of series-connected cells on each string of the array that characterizes the PV generator – see [14] and [29] for more details. The other two terms are the Boltzmann constant ($k = 1.38065 \times 10^{-23}$ J/K) and the absolute value of the electron charge ($q = 1.60218 \times 10^{-19}$ C).

Since I and V can be recorded at the PV generator terminals and T_c can be measured on its (back) surface [9], then (1) has *five characteristic parameters*: I_{ph} , I_0 , n (or a), R_{sh} , and R_s . In general, these parameters assume values within well-known ranges. For instance, I_{ph} usually approaches the value of the short-circuit current verified for the PV generator at the given operating condition [14]. In turn, I_0 typically assumes a low order of magnitude value, such as micro or nanoamperes [14]. Regarding n , its value essentially varies between 1 and 2 [14]; consequently, a assumes values of some units [27]. About R_{sh} , it can assume a value in the order of tens to thousands of ohms, while R_s hardly exceeds a few units of ohm [12].

Finally, it is crucial to note that the calculation of (1) cannot be straight carried out. The fact that the variable I is present inside and outside the exponential term prevents it from being isolated in this equation using algebraic operations alone – that is, $I = f(V, I)$. This condition makes the characteristic equation of the single diode model essentially *transcendental* concerning such a variable. Accordingly, some alternative mathematical approach must specifically be adopted to allow the determination of its solution.

2.2. Approach via the Lambert W function

The solutions of diverse transcendental equations can be expressed in terms of the *Lambert W function* [23]. Strictly speaking, this special function is defined as the inverse of the function given by $f(x) = x \exp(x)$, where x is any complex number [24]. Nevertheless, as demonstrated in [14], it satisfies the relationship delivered by (2) so that z is an image of $f(x)$ and $W\{z\} \equiv x$. Hence, the main advantage offered by the Lambert W function is that it allows the required value of x to be explicitly determined from the knowledge of just z .

$$z = W\{z\} \exp(W\{z\}) \quad (2)$$

Nonetheless, the use of the Lambert W function is a task that requires some care. That is because this function is essentially multivalued, having two branches in the real domain and infinite branches in the complex domain [30]. For instance, Fig. 2 shows its real branches. As can be seen, there are two possible real values for $W\{z\}$ when

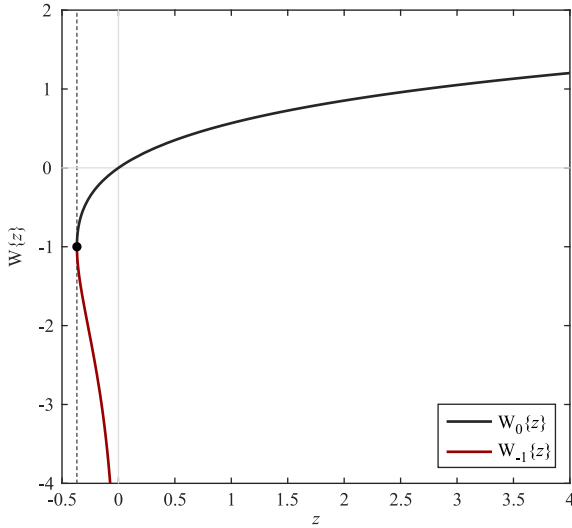


Fig. 2. Real branches of the Lambert W function.

$z \in [-\exp(1)^{-1}, 0)$, i.e., the Lambert W function is not injective into this interval [31]. Thus, the bottom branch, designated by W_{-1} , is defined for $z \in [-\exp(1)^{-1}, 0)$ with $W\{z\} \leq -1$ [23]. In turn, the upper branch, or the *principal branch*, is designated by W_0 and defined for $z \in [-\exp(1)^{-1}, \infty)$ with $W\{z\} \geq -1$ [23].

Aware of that, it is possible to manipulate the characteristic equation of the single diode model so that it assumes a similar form to (2). As a result, it is found that (1) can be rewritten as (3) [27].

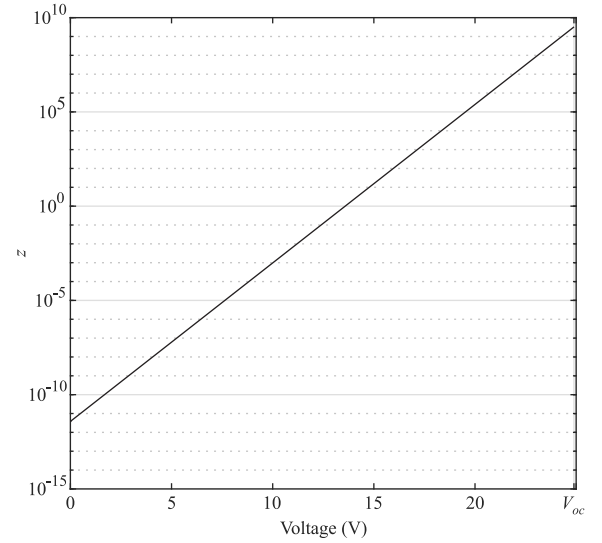
$$I = \frac{1}{R_{sh} + R_s} (R_{sh} (I_{ph} + I_0) - V) - \frac{a}{R_s} W_0\{z\} \quad (3a)$$

$$z = \frac{I_0 R_{sh} R_s}{a (R_{sh} + R_s)} \exp\left(\frac{R_{sh}}{a (R_{sh} + R_s)} (R_s (I_{ph} + I_0) + V)\right) \quad (3b)$$

In (3a), the Lambert W function must be evaluated on its principal branch because all terms involved in (3b) assume only non-negative real values at actual physical conditions [14,27]. Besides, it is important to note that the result of this latter equation can vary widely. Since R_{sh} can reach thousands of ohms and R_s nears zero, the value of z is predominantly influenced by the magnitude of V [27]. As illustrated in Fig. 3, for small values of V , z tends to assume values approaching zero. In contrast, for values of V close to the open-circuit voltage (V_{oc}), z tends to assume values that can even exceed the order of billions.

When analyzing (3), one can conclude that it allows the explicit calculation of I ; however, only if the correspondent value of the Lambert W function at the required point z is previously known. In other words, this equation translates $I = f(V, W_0\{z\})$. Therefore, the issue involved in this approach of the single diode model is no longer “how to calculate its characteristic equation?” but “how to calculate the Lambert W function at any non-negative point of its principal branch?”.

Last but not least, it is pertinent to mention that the approach via the Lambert W function of the single diode model also allows writing its characteristic equation focused on V , i.e., $V = f(I, W_0\{z\})$. Nonetheless, works such as those conducted in [17] and [27] indicate that such an alternative formulation allows simulating the I - V curve of a PV generator with less accuracy than (3). Furthermore, unlike this last equation, it is subject to arithmetic overflow due to the elevated order of magnitude of the values assumed by the Lambert W function argument [27]. For instance, the results reported in [27] reveal that the values of z can exceed the order of 10^{50000} on this alternative formulation; at the same time, a typical computational environment, based on the IEEE double precision floating point standard, has a representation limit of around 10^{308} [27]. Since the formulation focused on V has these drawbacks, the present work will only consider the

Fig. 3. Magnitude of z concerning the operating voltage of a PV generator.

one presented in (3) – which, by the way, appears to be the most widespread formulation in literature [14]. The formulation focused on V will be left to be addressed in more detail in future work.

3. Methodology

As introduced in Section 1, some methods have already been specifically developed to allow the calculation of the Lambert W function within the scope of the single diode model. Below, an overview of these methods is duly presented. Next, their respective strengths and weaknesses, as well as their improvement possibilities, are discussed. From the conclusions obtained, the proposed method is then designed.

3.1. An overview of related methods

It is observed in the literature that one of the first methods specifically developed for calculating the Lambert W function in the interval of interest of the single diode model was introduced in the work presented by [28]. Later, one of its authors called it the *hybrid method* [26]. Strictly speaking, this method combines two series expansions of the Lambert W function. The former, given by (4), consists of the first six terms of the series expansion proposed in [24] and is applied to $z \in [0, 9]$ [26]. The latter, given by (5), consists of the first seven terms of the asymptotic expansion and is applied for values of $z \in [9, \infty]$ [26]. The results reported in [17] and [26] demonstrate that the hybrid method is significantly faster than the Matlab function “*lambertw*” and can return solutions to the single diode model with an uncertainty of the order of 10^{-5} A.

$$W_0\{z\} = u + \frac{u}{1+u}p + \frac{1}{2}\frac{u}{(1+u)^3}p^2 - \frac{1}{6}\frac{u(2u-1)}{(1+u)^5}p^3 + \frac{1}{24}\frac{u(6u^2-8u+1)}{(1+u)^7}p^4 - \frac{1}{120}\frac{u(24u^3-58u^2+22u-1)}{(1+u)^9}p^5 \quad (4a)$$

$$u = \frac{z}{\exp(1)} \quad (4b)$$

$$p = 1 - \frac{z}{\exp(1)} \quad (4c)$$

$$W_0\{z\} = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(-2+L_2)}{2(L_1)^2} + \frac{L_2(6-9L_2+2(L_2)^2)}{6(L_1)^3}$$

$$+ \frac{L_2(-12 + 36L_2 - 22(L_2)^2 + 3(L_2)^3)}{12(L_1)^4} + \frac{L_2(60 - 300L_2 + 350(L_2)^2 - 125(L_2)^3 + 12(L_2)^4)}{60(L_1)^5} \quad (5a)$$

$$L_1 = \ln(z) \quad (5b)$$

$$L_2 = \ln(\ln(z)) \quad (5c)$$

More recently, the work conducted on [27] presents an interesting strategy to calculate the Lambert W function. Working on different parameterizations of this function, its authors proposed using the sets of mathematical expressions given by (6) and (7) for its calculation [27]. Note that the value of the auxiliary variable s in (6a) is given by the root that satisfies (6b). Likewise, the value of the auxiliary variable w in (7a) is given by the root that satisfies (7b). The authors suggest using the Newton–Raphson method to calculate these roots, considering respectively (6c) and (7c) as initial guesses [27]. Furthermore, the choice for one of these sets obeys a conditional logic similar to that of the hybrid method. More precisely, (6) is applied for $z \in [0, \exp(1)]$ and (7) for $z \in \exp(1), \infty$ [27].

$$W_0\{z\} = zs \quad (6a)$$

$$s - \exp(-z)^s = 0 \quad (6b)$$

$$s^0 = \frac{1 + \ln(1 - \ln(\exp(-z)))}{1 - 2\ln(\exp(-z))} \quad (6c)$$

$$W_0\{z\} = w \quad (7a)$$

$$\ln(w) + w - \ln(z) = 0 \quad (7b)$$

$$w^0 = \ln(z) - \ln(\ln(z)) + \frac{\ln(\ln(z))}{\ln(z)} + \frac{\ln(\ln(z))(-2 + (\ln(\ln(z))))}{2(\ln(z))^2} \quad (7c)$$

Following this strategy, two methods were then designed by [27]. The former, called the *numeric method*, allows an unrestricted number of iterations for internal applications of the Newton–Raphson method [27]. The latter, called the *explicit method*, does not allow any iteration for internal applications of the Newton–Raphson method [27]. In other words, once the choice is made between (6a) or (7a), an initial guess is duly calculated from (6c) or (7c), and a single execution of the characteristic equation of the Newton–Raphson method is performed on (6b) or (7b) in this method. The reported results confirm that the explicit method is faster than the numeric method and reveal that its performance is comparable to that of the hybrid method [27]. In contrast, these results indicate that the numeric method can return solutions to the single diode model with an uncertainty of the order of 10^{-16} A at a simulation time of about three orders of magnitude less than the corresponding Matlab function [27].

3.2. Preliminary discussion

From a practical point of view, these three methods presented above have pros and cons. At first glance, the hybrid and explicit methods draw attention for being non-iterative and, thereby, having a fixed computational cost. However, their level of accuracy is a point that deserves awareness. Considering the IEEE double precision floating point standard, the order of *machine precision*, i.e., the upper limit of the relative approximation error due to rounding in floating point arithmetic, is about 2.2×10^{-16} [32]. Accordingly, the solutions returned by these methods are of intermediate quality and may even be inappropriate for certain studies. For instance, the results presented in [17] demonstrate that the solutions found for the traditional problem

of the RTC France cell via the hybrid method are inferior to those found via the corresponding Matlab function and, therefore, ineffective.

In turn, the numeric method stands out for its potential to achieve solutions so accurately that they result in an uncertainty of the order of machine precision. Despite that, the reported results make it clear that more than one iteration of the Newton–Raphson method is typically necessary – otherwise, the explicit method would also return solutions with such a high level of quality. So, this method has a variable computational cost that tends to be higher than those verified for the non-iterative methods.

Not enough, it is important to note that both methods have a common sensitive point: the conditional nature. That is, the fact that the mathematical expressions applicable within the coverage of each method depend on the value of z is a characteristic that burdens their computational implementations with “if-else” statements.

From this assessment, it is possible to conclude that a novel method to be proposed for calculating the Lambert W function in the interval of interest of the single diode model should ideally be capable of returning solutions with a level of accuracy (and, consequently, of uncertainty) of the order of machine precision, at the lowest possible computational cost and, preferably, fixed.

Designing a method that meets these characteristics based on the logic adopted by existing methods is not a promising task. For example, improving the accuracy of the hybrid method would imply expanding each series in additional terms or even considering new series expansions for certain intervals. Both options would make its implementation more complicated and increase its computational cost. Along the same lines, improving the performance of the explicit and numeric methods would imply replacing the Newton–Raphson method with a root-finding algorithm that would be more efficient and, consequently, more complex. The search for more accurate initial guesses would also be difficult since it would have to be done on two fronts, one for each considered parametric version of the Lambert W function. Furthermore, none of these mentioned possibilities for improvement would eliminate the conditional nature of these methods.

Thus, a broad investigation was carried out in the specific literature in order to find a more attractive logic for designing a novel method for calculating the Lambert W function in the interval of interest of the single diode model. As a result, it was found that the logic of a *predictor–corrector method for solving nonlinear equations* would allow the development of a method that met the desired characteristics [33,34].

3.3. Proposed method

In general, a predictor–corrector method is built from the association of two methods already proposed for the same purpose, aiming to obtain a new, in some way more efficient [32–34]. In the context of solving nonlinear equations, it typically starts with an arbitrary initial guess and then carries out two chained steps to improve its accuracy [33,34]. The first step predicts an approximate solution for the Lambert W function using the characteristic equation of one of the selected methods. The second, in turn, corrects some decimal places of this approximate solution using the characteristic equation of the other selected method. If the solution found in the second step is not accurate enough, it is then used as a new initial guess for the first step, and this entire process is repeated until a solution with the required level of quality is reached.

Aware of that, a predictor–corrector method was particularly built for calculating the Lambert W function in the interval of interest of the single diode model. Aiming to ensure its application is self-contained and achieves solutions with the desired level of accuracy in a non-iterative manner, an equation was precisely adopted to determine its initial guesses. The details of this *self-started predictor–corrector method* are carefully presented below. Ultimately, its leading equations are summarized, and its computational implementation is duly elucidated.

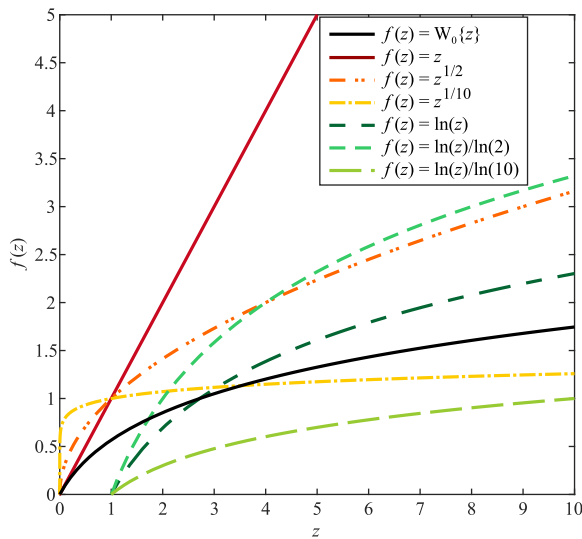


Fig. 4. Comparison of the Lambert W function with other regular functions, taking into account the interval of interest of the single diode model.

3.3.1. Start step

Finding an approximation, yet rough, for the non-negative real interval of the Lambert W function is a challenging task. As shown in Fig. 4, this interval has a singular nonlinear nature.

To deal with it, the methods presented in Section 3.1, as well as the majority of methods developed for calculating the Lambert W function within other scopes, adopt a strategy of “divide et impera” [24,27,28,30,31,35]. In other words, the non-negative interval of the principal branch is split into a few subintervals, and a mathematical expression is particularly assigned to represent each one approximately. Although some of these strategies present surprising levels of accuracy (see [31], for example), the complexity of their equations and conditional natures make their computational implementation costly to the calculation process of the Lambert W function.

In contrast, it is observed in the specific literature that some non-conditional formulations have already been constructed to represent the mentioned interval of interest [36]. For instance, the work described in [37] presents a simple equation for this purpose. It is given by (8), being $W_0\{z\} \approx W_i$. Note that the subscript “i” of W_i refers to the “initial”.

$$W_i = \ln \left[1 + \frac{z}{1 + 0.5 \ln(1 + z)} \right] \quad (8)$$

According to the work conducted in [36], (8) can deliver approximations for the Lambert W function with the relative error limit of 3.53×10^{-2} for $z \in [0, \infty)$. This level of accuracy is not impressive but is reasonable for an initial guess. So, judging by its low complexity and fittingness, such an equation was chosen to start the proposed method.

3.3.2. Predictor step

In order to improve the quality of the initial approximation, the very idea of calculating the Lambert W function can be exploited. To be clearer, consider that the initial guess delivered by (8) differs from the exact solution $W_0\{z\}$ by an error denoted by δ_i . Mathematically, this translates to (9).

$$W_0\{z\} = W_i + \delta_i \quad (9)$$

When applying (9) to (2) and executing some manipulations, it is found that such an error can be determined through (10). If this equation could be strictly satisfied, then the initial guess delivered by (8) could be corrected immediately. However, it has a transcendental nature – since δ_i is outside and inside the last logarithm term on the

right-hand side. So, its calculation cannot be carried out using algebraic operations alone.

$$\delta_i = \ln \left(\frac{z}{W_i} \right) - W_i - \ln \left(1 + \frac{\delta_i}{W_i} \right) \quad (10)$$

To overcome this situation, the work developed in [37] suggests expanding the logarithmic term in question into a Taylor series. When considering this series expansion, truncated in its first term, the relationship shown in (11) is obtained.

$$\ln \left(1 + \frac{\delta_i}{W_i} \right) \approx \frac{\delta_i}{W_i} \quad (11)$$

By substituting (11) into (10) and executing some manipulations, it is found that the error associated with W_i is approximately given by (12).

$$\delta_i \approx \frac{W_i}{1 + W_i} \left[\ln \left(\frac{z}{W_i} \right) - W_i \right] \quad (12)$$

By merging (9) and (12), it is found that $W_0\{z\} \approx W_p$, where W_p is given by (13). This relatively simple equation is the basis of the recursive method proposed by [37] for calculating the Lambert W function. According to its authors, it is capable of making a second-order correction [37]. In other words, the number of correct decimal places of the initial guess approximately doubles with its execution.

$$W_p = \frac{W_i}{1 + W_i} \left[1 + \ln \left(\frac{z}{W_i} \right) \right] \quad (13)$$

Corroborating this, the work conducted in [36] points out that the combination of (8) and (13) can yield candidate solutions for the Lambert W function with the relative error limit of 2.16×10^{-4} for $z \in [0, \infty)$. These candidate solutions still have a low level of accuracy; however, it is important to draw attention again to the low complexity of the equations that yield them. Therefore, (13) was chosen to compose the predictor step of the proposed method.

3.3.3. Corrector step

The corrector step must be executed immediately after the predictor step. Thus, (10) takes the form shown by (14).

$$\delta_p = \ln \left(\frac{z}{W_p} \right) - W_p - \ln \left(1 + \frac{\delta_p}{W_p} \right) \quad (14)$$

As discussed, the Taylor series expansion of the last logarithm term on the right-hand side can produce a second-order corrector equation. Although its simplicity is a great attraction, other alternatives, more elaborated but with superior performance, can be considered opportunely now. For instance, the logarithm term related to δ_p can be appropriately expanded in a continued fraction [35]. So, when considering this series expansion, truncated in its two first terms, (15) is obtained.

$$\ln \left(1 + \frac{\delta_p}{W_p} \right) \approx \frac{\delta_p W_p + \frac{1}{6} (\delta_p)^2}{(W_p)^2 + \frac{2}{3} \delta_p W_p} \quad (15)$$

By substituting (15) into (14) and executing some manipulations, the quadratic equation described in (16) is found. Note that the auxiliary parameter y , specified by (16b), was isolated from (16a) to clarify its visualization.

$$\underbrace{\left(\frac{2}{3} W_p + \frac{1}{6} \right) \delta_p^2}_A + \underbrace{\left(W_p^2 + W_p - \frac{2}{3} y W_p \right) \delta_p}_B + \underbrace{\left(-y W_p^2 \right)}_C = 0 \quad (16a)$$

$$y = \ln \left(\frac{z}{W_p} \right) - W_p \quad (16b)$$

The error associated with W_p is defined by one of the roots of (16). Nonetheless, finding the roots of a quadratic equation is a sensitive issue from the perspective of numerical computation [38]. When B^2 is expressively larger than $4AC$, the traditional *quadratic formula*, given by (17), is subject to subtractive cancellation and, thereby, the loss of

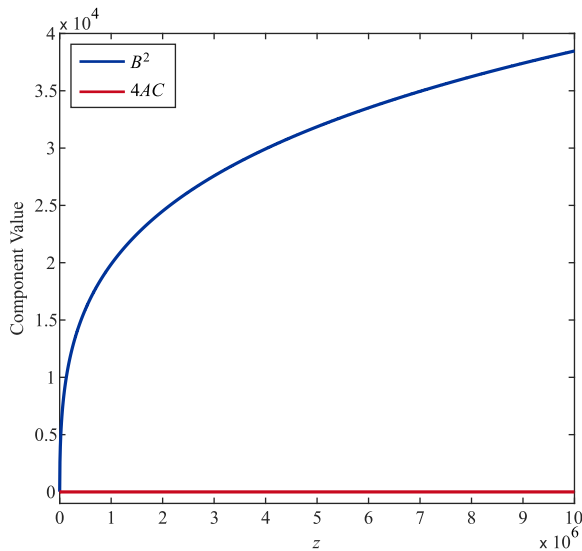


Fig. 5. Relationship between B^2 and $4AC$ verified for the characteristic equation of the single diode model when employed to represent a PV generator.

correct significant digits of the solution [32]. By the way, Fig. 5 shows this is precisely the case verified in the scope of the single diode model.

$$\delta_p = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \quad (17)$$

As an alternative to the traditional quadratic formula, several works in the literature recommend using the *citardauq formula* (the reverse of the “quadratic” formula), which is computationally more stable [32, 38]. For the present case, it is given by (18).

$$\delta_p = \frac{-2C}{B \pm \sqrt{B^2 - 4AC}} \quad (18)$$

It is pertinent to clarify that the plus-minus sign ($\pm$) present in the denominator of (18), as well as in the numerator of (17), is due to the compact representation of the two roots of the quadratic equation given by (16). That is, the former root is obtained by considering the positive sign, and the latter by considering the negative sign. Thus, by solving (16) through the root obtained considering the positive sign in (18) and executing some manipulations, the mathematical expression given in (19) is found. It is worth highlighting that such a root is considered to the detriment of the one obtained with a negative sign because it tends to 0 as y tends to 0 and, thereby, dignifies the expected behavior [35]. Moreover, the root obtained with a negative sign can easily explode to infinite, which is numerically inappropriate.

$$\delta_p = \frac{2yW_p}{\left(1 + W_p - \frac{2}{3}y\right) + \sqrt{\left(1 + W_p + \frac{2}{3}y\right)^2 - 2y}} \quad (19)$$

From (19), it is then possible to write a corrector equation as demonstrated by (20). This equation is the core of the recursive method developed in [35] for calculating the Lambert W function. If, on the one hand, it is more complex than (13), on the other hand, it is capable of making a fourth-order correction [31]. That means the number of correct decimal places of a candidate solution yielded in the predictor step is approximately quadruplicated with its execution.

$$W_c = W_p + \delta_p \approx W_p + \frac{2yW_p}{\left(1 + W_p - \frac{2}{3}y\right) + \sqrt{\left(1 + W_p + \frac{2}{3}y\right)^2 - 2y}} \quad (20)$$

Taking into account that W_p has a relative error limit of 2.16×10^{-4} for $z \in [0, \infty)$, then W_c addresses a relative error limit of the order of 10^{-16} for the same interval. It is relevant to remember that in the

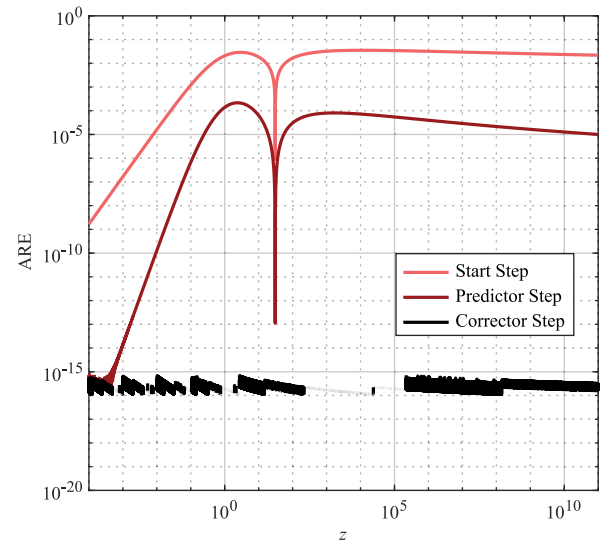


Fig. 6. ARE profile verified in each step of the proposed method.

IEEE double precision floating point standard, the machine precision is about 2.2×10^{-16} . Therefore, the solution pointed out by W_c in the first execution of the “start step–predictor step–corrector step” sequence can be considered effective to $W_0\{z\}$ – i.e., $W_0\{z\} = W_c$ – with no iteration of the predictor–corrector steps being necessary.

3.3.4. Consolidation of the method

Given the exposed content, the proposed method is summarized in (21). Note that (21a), (21b), (21c), and (21d) must be calculated sequentially. Nevertheless, there is no technical impediment to condensing these equations. If the user wishes, this can be done even to the point where they compose just one. This form of presentation was chosen to be more friendly.

$$W_i = \ln \left[1 + \frac{z}{1 + 0.5 \ln(1 + z)} \right] \quad (21a)$$

$$W_p = \frac{W_i}{1 + W_i} \left[1 + \ln \left(\frac{z}{W_i} \right) \right] \quad (21b)$$

$$y = \ln \left(\frac{z}{W_p} \right) - W_p \quad (21c)$$

$$W_c = W_p + \frac{2yW_p}{\left(1 + W_p - \frac{2}{3}y\right) + \sqrt{\left(1 + W_p + \frac{2}{3}y\right)^2 - 2y}} \quad (21d)$$

As explained, this method guarantees solutions for the Lambert W function with a level of accuracy of the order of machine precision. In order to illustrate this feature, Fig. 6 presents the *absolute relative error* (ARE) profile of each of its steps, considering non-negative real arguments for the Lambert function. It is worth noting that this error metric was computed according to (22), where $W_0\{z\}_{calc}$ is the solution calculated for the Lambert W function at a required point z using a specific step of the proposed method, i.e., using (21a) in the case of start step, (21a) and (21b) in the case of the predictor step, and (21a), (21b), (21c), and (21d) in the case of the corrector step. In turn, $W_0\{z\}_{ref}$ is the respective benchmark solution, then calculated using the Matlab function “*lambertw*”. Note that the smaller the ARE value, the more accurate the calculated solution is.

$$ARE(z) = \left| \frac{W_0\{z\}_{calc} - W_0\{z\}_{ref}}{W_0\{z\}_{ref}} \right| \quad (22)$$

As can be seen, the ARE values verified for the start and predictor steps do not exceed the relative error limits previously documented. Besides, it is essential to observe that the ARE values verified for the

Table 1
Parameters of the case studies [27].

Parameters	Case study #1	Case study #2	Case study #3	Case study #4	Case study #5	Case study #6
I_{ph} (A)	15.8800	1.0320	3.6540	0.5780	0.7610	4.8020
I_0 (A)	7.4400E-10	2.5130E-06	3.9990E-21	1.3400E-10	3.1070E-07	4.0160E-07
a (V)	14.6700	1.3000	0.5160	0.0118	0.0390	0.0370
R_{sh} (Ω)	425.2000	744.7140	2329.0000	612.0000	52.8900	1167.0000
R_s (Ω)	2.0400	1.2390	2.6900	0.0127	0.0370	0.5906
V_{oc} (V)	348.1000	16.7750	24.8930	0.2620	0.5730	0.6030

corrector step are, as envisioned, always in an order of magnitude lower than 10^{-15} .

In addition to the proposed method's high level of accuracy, it is important to highlight that the self-starting promoted by (21a) and the moderate complexity of (21b), (21c), and (21d) make its use convenient. Also, its non-conditional and non-iterative nature makes its computational cost relatively low and fixed. Consequently, a short simulation time is expected.

That said, Algorithm 1 describes a pseudo-code for the computational implementation of the proposed method, then called the *self-started predictor-corrector method*, within the scope of the single diode model. As can be noticed, this is a fairly easy task.

Algorithm 1 Calculation of the characteristic equation of the single diode model using the proposed method.

Input: I_{ph} , I_0 , a , R_{sh} , R_s , V

Output: I

- 1: Define I_{ph} , I_0 , a , R_{sh} , R_s , V
- 2: Determine z via (3b)
- 3: Calculate W_i via (21a)
- 4: Calculate W_p via (21b)
- 5: Calculate y via (21c)
- 6: Calculate W_c via (21d)
- 7: Calculate I via (3a), considering $W_0\{z\} = W_c$
- 8: **Return** I

4. Experimental results

In order to validate the proposed method, the six case studies recently presented by [27] were taken as a basis. Each of them consists of an I - V curve, which is generated from the respective data presented in Table 1 [27]. In this process, the corresponding values of I_{ph} , I_0 , a , R_{sh} , and R_s are substituted in (3), and an “ I ” is particularly calculated for each “ V ” considered between zero and V_{oc} . It is worth mentioning that the values assumed by z in these cases vary between 10^{-12} and 10^{35} and, thereby, comprise a fairly representative range [27].

For comparison purposes, the I - V curve of each case study was initially generated using the proposed method and the recursive methods selected for its building, namely, the second-order method proposed by [37] and the fourth-order method proposed by [35]. It is worth noting that in both cases, the initial guesses were delivered by (21a). Subsequently, the methods presented in Section 3.1 were also used, namely, the hybrid [26], the explicit [27], and the numeric methods [27].

All these methods were self-implemented in Matlab software, version 2016b. Thus, the I - V curves obtained through its function “*lambertw*” were taken as a benchmark. It is also important to mention that its function “*linspace*” was used to specify $N = 1000$ equally spaced points between 0 and V_{oc} for each I - V curve.

The related simulations were executed on a PC with an Intel Core i7-4510U @ 2/2.60 GHz, 8 GB RAM, under the Windows 10 64-bit OS. The found results are objectively reported below.

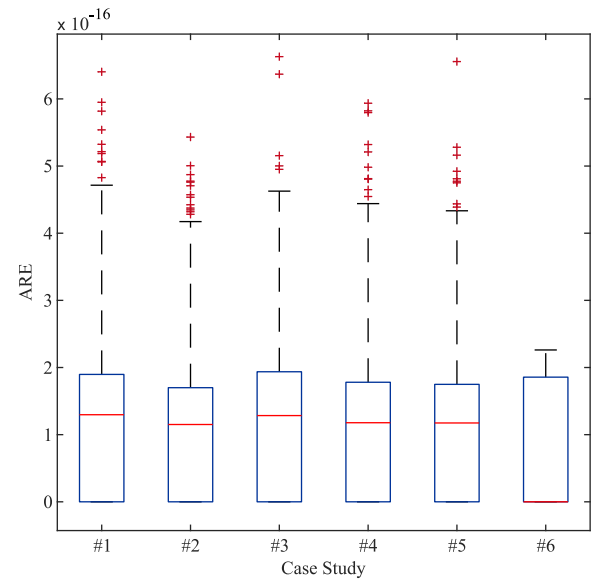


Fig. 7. Boxplot of ARE values verified for the proposed method.

4.1. Goal accomplishment

As explained in the previous section, the proposed method for calculating the Lambert W function in the interval of interest of the single diode model was developed to return solutions with a level of accuracy on the order of machine precision and at a low computational cost. In order to check whether it accomplishes its goal, two evaluation metrics were adopted: ARE and total mean simulation time.

The ARE was computed according to (22). Accordingly, Fig. 7 depicts a statistical overview of the ARE values verified for the proposed method in each case study. As can be seen, all of them are of the order of 10^{-16} . Better than that, the third quartiles are below 2×10^{-16} in all case studies. So, at least 75% of the ARE values are less than the machine precision. Furthermore, it is interesting to note that, particularly in case study #6, the median of the ARE values approaches the minimum one (this situation is later detailed in Table 2).

The total mean simulation time, in turn, was computed by applying 100 times the proposed method in each case study (such as in [17]). The solutions found were obviously the same in every application, but the respective simulation times slightly differed due to the PC's computational instabilities. Hence, the mean of these times was recorded to obtain a more consistent evaluation metric.

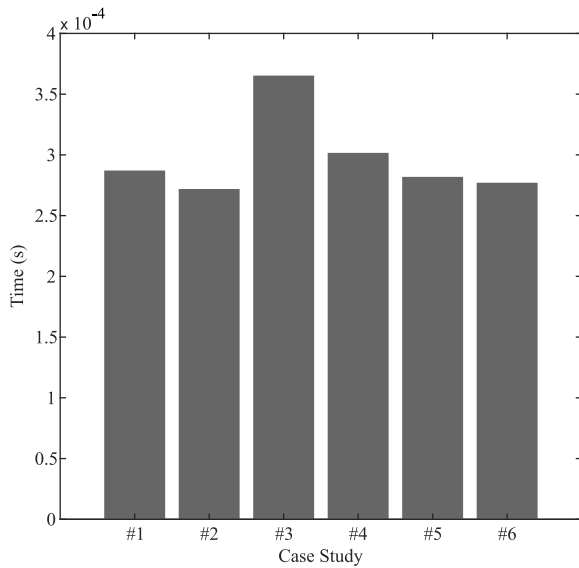
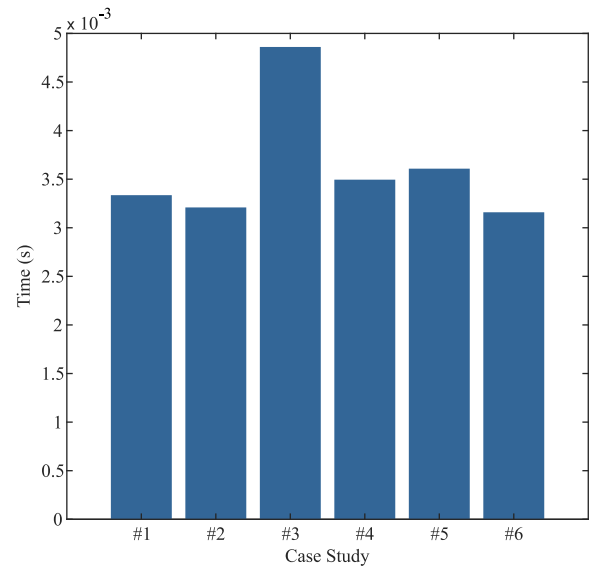
That said, Fig. 8 shows the total mean simulation time spent by the proposed method to calculate all values of $W_0\{z\}$ required in each case study. As can be seen, these times are not expressive. Considering that each comprises 1000 calculations of the Lambert W function, it is possible to infer that the proposed method requires only around 10^{-7} seconds to calculate a single $W_0\{z\}$.

Additionally, Fig. 9 shows the total mean simulation time spent to perform all required calculations of the characteristic equation of the single diode model using the proposed method. As expected, these times also have a significantly reduced order of magnitude. Not enough,

Table 2

Main statistical metrics of the ARE values verified for the proposed method and the recursive methods selected for its building.

Case study	Alternative	Minimum	Median	Mean	Maximum	Standard deviation
#1	2nd-order method	0.0000E+00	1.4775E-16	1.5055E-16	6.7588E-16	1.3323E-16
	4th-order method	0.0000E+00	1.2806E-16	1.2080E-16	6.0963E-16	1.2078E-16
	Proposed method	0.0000E+00	1.2969E-16	1.2201E-16	6.4029E-16	1.2243E-16
#2	2nd-order method	0.0000E+00	1.3150E-16	1.2484E-16	6.3678E-16	1.2603E-16
	4th-order method	0.0000E+00	1.1469E-16	1.0327E-16	5.9449E-16	1.1890E-16
	Proposed method	0.0000E+00	1.1513E-16	1.0075E-16	5.4323E-16	1.1322E-16
#3	2nd-order method	0.0000E+00	1.3798E-16	1.3842E-16	6.6292E-16	1.3391E-16
	4th-order method	0.0000E+00	1.3094E-16	1.2054E-16	6.6292E-16	1.2025E-16
	Proposed method	0.0000E+00	1.2842E-16	1.1792E-16	6.6292E-16	1.1994E-16
#4	2nd-order method	0.0000E+00	1.4316E-16	1.4175E-16	6.4520E-16	1.3176E-16
	4th-order method	0.0000E+00	1.2220E-16	1.1311E-16	5.9359E-16	1.2184E-16
	Proposed method	0.0000E+00	1.1777E-16	1.0746E-16	5.9359E-16	1.2037E-16
#5	2nd-order method	0.0000E+00	1.3760E-16	1.3456E-16	8.1948E-16	1.2944E-16
	4th-order method	0.0000E+00	1.2151E-16	1.0782E-16	6.5558E-16	1.1437E-16
	Proposed method	0.0000E+00	1.1742E-16	1.0302E-16	6.5558E-16	1.1216E-16
#6	2nd-order method	0.0000E+00	0.0000E+00	5.0508E-17	4.4442E-16	8.3650E-17
	4th-order method	0.0000E+00	0.0000E+00	5.8279E-17	2.3048E-16	8.6816E-17
	Proposed method	0.0000E+00	0.0000E+00	6.1373E-17	2.2613E-16	8.8500E-17

**Fig. 8.** Total mean simulation times spent by the proposed method on the required calculations of the Lambert W function.**Fig. 9.** Total mean simulation times spent on the required calculations of the characteristic equation of the single diode model, considering the solutions returned by the proposed method.

it is pertinent to note that these times of the order of 10^{-3} seconds are suitable even for critical studies, such as those related to the parameter extraction problem of the single diode model.

These results ascertain, therefore, that the proposed method accomplishes its goal of returning solutions with a level of accuracy on the order of machine precision for the Lambert W function in the interval of interest of the single diode model at a low computational cost.

4.2. Development pertinence

Although the proposed method accomplishes its goal, it is also necessary to check out the quality of its performance regarding the second-order and fourth-order recursive methods selected for its building. Otherwise, its materiality may be called into question.

In this sense, Table 2 describes the main statistical metrics of the ARE values verified for the solutions returned by the proposed method, as well as for the second-order and fourth-order recursive methods. The best comparative results are highlighted in **boldface**.

As can be noticed, the ARE values verified for the proposed method are similar to those verified for the other two methods. Given their reduced orders of magnitude, it is possible to state that the proposed method preserves the high level of accuracy, robustness, and stability of the recursive methods selected for its building.

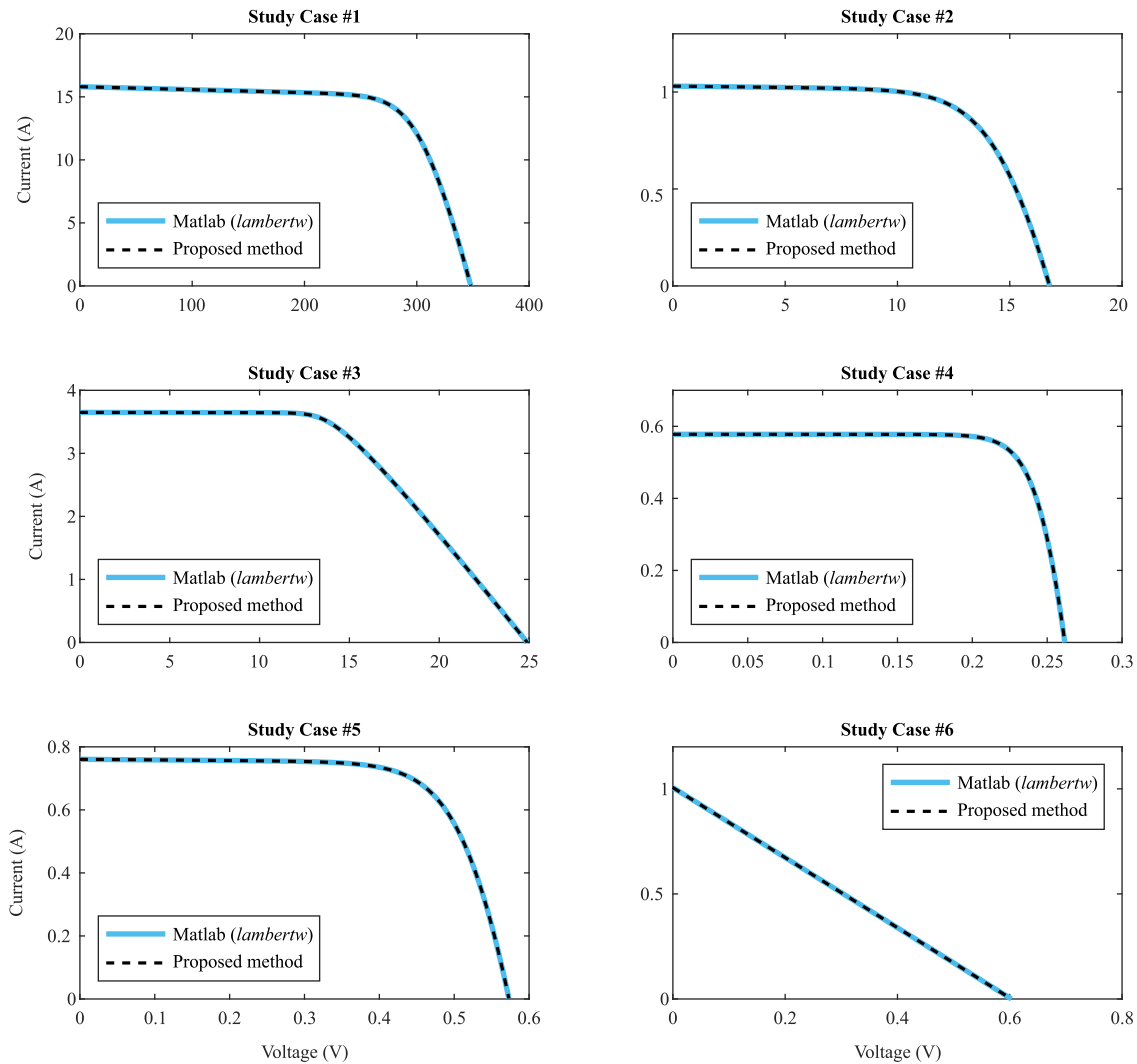
In turn, Table 3 presents a qualitative comparison of the total mean simulation time spent by the proposed method to calculate all values of $W_0\{z\}$ required in each case study regarding those recorded for the second-order and fourth-order recursive methods. More specifically, the total mean simulation time recorded for each alternative is normalized concerning that recorded for the proposed method in the respective case study (see Fig. 8).

These results reveal that the proposed method is at least two times faster than the second-order and fourth-order recursive methods. Consequently, they prove that its predictor–corrector structure meets the premise of being more efficient than that of the recursive methods

Table 3

Comparison of the total mean simulation times recorded for the required calculations of the Lambert W function.

Alternative	Case study #1	Case study #2	Case study #3	Case study #4	Case study #5	Case study #6
2nd-order method	2.0759	2.2258	2.1787	2.6684	2.1408	2.5291
4th-order method	2.2588	2.4139	2.4038	2.4657	2.2808	2.5302
Proposed method	1	1	1	1	1	1

**Fig. 10.** Comparison of the I - V curves generated for each case study from the Matlab function “*lambertw*” and the proposed method.

selected for its building. Likewise, they confirm the pertinence of its development.

4.3. Competitiveness

Having proven the success of the development of the proposed method, it is now necessary to check out its competitiveness compared to other potential alternatives. In general, the *root mean square error* (RMSE) is adopted as a figure of merit to evaluate the effectiveness of applying the methods for calculating the Lambert W function within the scope of the single diode model [17,26,27]. Thus, it is calculated as demonstrated by (23), where $I_{calc,j}$ is the current value calculated for the point “ j ” of voltage using a method in question, and $I_{ref,j}$ is the respective value calculated using the Matlab function “*lambertw*”. Note

that the smaller the RMSE value, the lower the uncertainty associated with the I - V curve generated from the solutions returned by such a method.

$$RMSE = \sqrt{\frac{\sum_{j=1}^N [I_{calc,j}(V_j) - I_{ref,j}(V_j)]^2}{N}} \quad (23)$$

That said, Table 4 presents the RMSE values verified for the proposed method in each case study. Additionally, the RMSE values verified for other alternatives already validated in the literature, such as the hybrid, explicit, and numeric methods, are provided.

As can be noticed, the RMSE values verified for the proposed method are equal or inferior to the order of machine precision in all cases. Consequently, it is observed in Fig. 10 that the I - V curves generated from its solutions are identical to those generated from the

Table 4
RMSE values verified for the proposed method and the related ones (in amperes).

Alternative	Case study #1	Case study #2	Case study #3	Case study #4	Case study #5	Case study #6
Hybrid method	7.0380E-04	1.1447E-05	1.8836E-05	7.7203E-06	1.0772E-05	3.9930E-11
Explicit method	4.8794E-04	1.0502E-05	9.0955E-06	2.1375E-07	8.2638E-07	8.3943E-16
Numeric method	6.9822E-16	3.3307E-17	3.1512E-16	1.8073E-17	2.3288E-17	5.3989E-16
Proposed method	6.7056E-16	2.9997E-17	2.8375E-16	1.5799E-17	2.0396E-17	4.6555E-16

Table 5
Comparison of the total mean simulation times recorded for the required calculations of the characteristic equation of the single diode model.

Alternative	Case study #1	Case study #2	Case study #3	Case study #4	Case study #5	Case study #6
Matlab (<i>lambertw</i>)	6021.1989	6668.1898	4090.1743	5461.3339	5948.2197	7022.3943
Hybrid method	1.3247	1.3835	1.2393	1.7322	1.2252	1.3628
Explicit method	1.0515	1.1123	0.9915	1.1319	1.0796	1.0069
Numeric method	1.2382	1.4121	1.2985	2.0333	1.1757	1.3144
Proposed method	1	1	1	1	1	1

corresponding Matlab function. Hence, these results corroborate that the proposed method can be effectively applied within the scope of the single diode model.

Concerning the other methods, the results in Table 4 reveal that the proposed method performs better than the non-iterative ones. For instance, the RMSE values verified for the hybrid method do not reach the order of 10^{-12} , and those verified for the explicit method only beat this order of magnitude in one of the six case studies. Furthermore, these results indicate that the solutions returned by the proposed method are comparable to those returned by the numeric method, which is iterative.

In addition, Fig. 11 presents the ARE profiles verified for the considered methods with respect to the I - V curve of each case study. As can be seen, the ARE values related to the solutions returned by the hybrid method are expressively higher than those verified for the other methods. Regarding the explicit method, it is observed in the first five case studies that their respective ARE values oscillate so that they rise some order of magnitude from a certain point of voltage and, consequently, from a certain value of z . In the sixth and last case study, it is observed that its ARE values stay closer to those verified for the numeric and proposed methods. About these last two methods, it is noted that their respective ARE values remain consistently on an order of magnitude equal to or less than that of the machine precision. So, these results substantiate and help understand the RMSE values in Table 4.

In turn, Table 5 presents a qualitative comparison of the total mean simulation time spent by the proposed method to calculate all values of $I_{calc,j}$ required in each case study regarding those recorded for the related methods and the corresponding Matlab function. As in Table 3, the total mean simulation time recorded for each alternative is normalized concerning that recorded for the proposed method in the respective case study (see Fig. 9). It is also worth remembering that each method was applied 100 times in each case study.

These results indicate that the proposed method is up to two times faster than the numeric method and tends to have a simulation time similar to or up to about 40% shorter than those recorded for the explicit and hybrid methods. Moreover, these results indicate that the proposed method is three orders of magnitude faster than the corresponding Matlab function.

Therefore, the proposed method is not only competitive but a particularly attractive alternative for calculating the Lambert W function interval of interest of the single diode model. First, because its effectiveness is comparable to the Matlab function "*lambertw*", but its efficiency is significantly higher. Second, its performance is superior to other methods specifically developed to meet the same purpose.

5. Conclusion

This paper proposed a self-started predictor-corrector method for calculating the Lambert W function in the interval of interest of the single diode model. As detailed, this method was carefully designed to return high-quality solutions at a low and fixed computational cost. In this sense, the moderate complexity of its four equations stands out, as well as its non-conditional and non-iterative nature.

In order to validate the proposed method, six appropriate case studies were taken as a basis. The results prove that it accomplishes its goal, and, thereby, it is capable of returning solutions for the Lambert W function with a relative error in the order of machine precision at a reduced simulation time. Consequently, the I - V curves generated by the single diode model from its solutions proved to be representative. Furthermore, these results demonstrated that the proposed method performs better than other state-of-the-art alternatives. In particular, it is more effective than the hybrid and explicit methods and significantly faster than the numeric method and the Matlab function "*lambertw*".

Accordingly, the self-started predictor-corrector method emerges as a reliable, easy-to-implement, and useful alternative for making the employment of the single diode model in studies more efficient. For future works, it would be engaging to assess its use considering the characteristic equation of this model focused on V . It would also be enjoyable to use it in studies related to the parameter extraction problem of the single diode model. In addition, it is opportune to apply such a method to other topics, which involve calculating the Lambert W function in the non-negative interval of its principal branch.

CRedit authorship contribution statement

Lucas Meirelles Pires Deotti: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ivo Chaves da Silva Junior:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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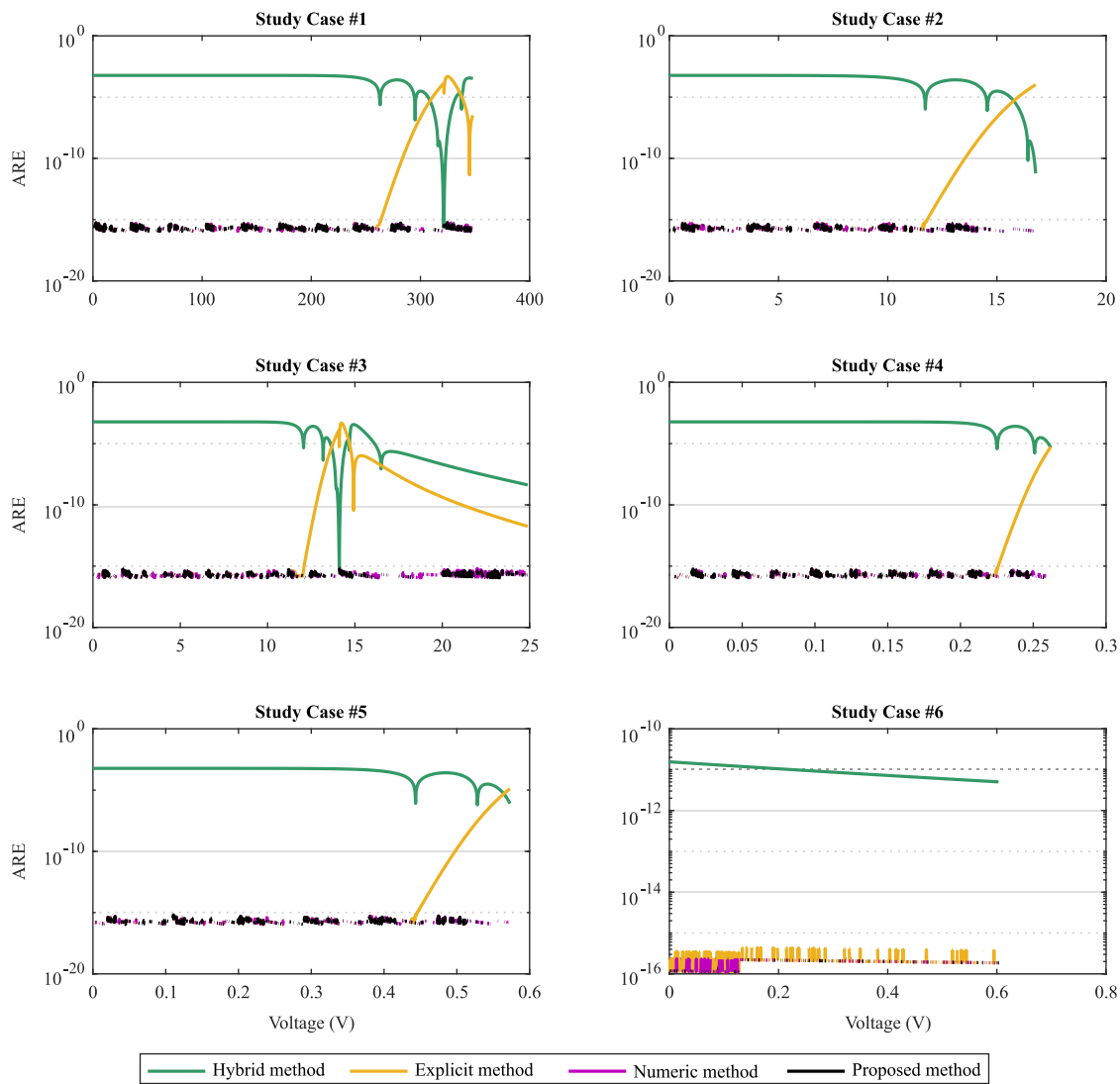


Fig. 11. Comparison of the ARE values verified for the considered methods in the I - V curve of each case study.

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